

London Underground “What-If” Simulation

User Stories Document

US-01 – View the Underground Network Map

As a **user**, I want to view an interactive map of the London Underground so that I can understand how stations and lines are connected.

Acceptance Criteria:

- AC1: The map displays all Underground lines with correct geographical alignment.
- AC2: Each line uses official TfL colours.
- AC3: Users can pan, zoom, and interact with the map smoothly.
- AC4: Stations are clearly visible, labelled, and selectable.

Priority: Must Have

US-02 – Accurate Station Hierarchy Representation

As a **transport analyst**, I want stations grouped into hubs with platforms and entrances, so the simulation reflects real Underground structure.

Acceptance Criteria:

- AC1: Parent stations represent multi-entrance/multi-platform hubs.
- AC2: Child nodes represent individual platforms or entrances.
- AC3: Node relationships match TfL StopPoint and line topology data.
- AC4: No orphan or disconnected station appears in the network graph.

Priority: Must Have

US-03 – Calculate the Shortest Route Between Stations

As a **commuter**, I want to calculate the shortest route between two stations so I can find the fastest journey.

Acceptance Criteria:

- AC1: Users can select a valid start and destination station.
- AC2: The system calculates the optimal path via Dijkstra/A*.
- AC3: Travel time and interchange information is displayed.
- AC4: Routes automatically update when network conditions change.

Priority: Must Have

US-04 – What-If Scenario Editor (Station & Line Closures)

As a **planner**, I want to manually close stations or lines so that I can analyse the impact of hypothetical disruptions.

Acceptance Criteria:

- AC1: Users can flag stations as “Open” or “Closed”.
- AC2: Closed stations are excluded from journey planning.
- AC3: Users can suspend entire lines or individual segments.
- AC4: Route calculations update after changes.
- AC5: The map visually reflects all simulated disruptions.

Priority: Must Have

US-05 – Live TfL Data Integration

As a **user**, I want the system to respond to real-time TfL service updates so that journey planning remains accurate.

Acceptance Criteria:

- AC1: Live disruptions from TfL API automatically update the network.
- AC2: Disrupted or closed lines/stations appear visually highlighted.
- AC3: Routing excludes live-closed stations automatically.
- AC4: Updates refresh dynamically without requiring page reload.

Priority: Must Have

US-06 – Live vs Hypothetical Mode Toggle

As a **user**, I want to toggle between Live and Hypothetical modes so I can compare real-world routes with simulated disruptions.

Acceptance Criteria:

- AC1: A clearly visible toggle switches between modes.
- AC2: Live mode reflects true TfL data; Hypothetical applies simulated edits.
- AC3: Switching modes recalculates routes in under 2 seconds.
- AC4: UI clearly indicates active mode (colour, tag, or banner).

Priority: Must Have

US-07 – Interactive Map Feedback for Disruptions

As a **user**, I want visual cues showing disruptions so I can understand their impact instantly.

Acceptance Criteria:

- AC1: Closed stations show a clear icon/strike-through.
- AC2: Suspended lines appear greyed out or dashed.
- AC3: Congestion and delays show colour gradients (if applicable).
- AC4: Map remains responsive and interactive.

Priority: Must Have

US-08 – Journey Metrics Display

As a **commuter**, I want journey metrics displayed so I can evaluate route quality.

Acceptance Criteria:

- AC1: Metrics include travel time, number of interchanges, and path breakdown.
- AC2: Metrics update in real time in both Live and Hypothetical modes.
- AC3: UI presents metrics clearly and consistently.

Priority: Must Have

US-09 – Simulate Crowding & Capacity Constraints

As a **transport planner**, I want to simulate overcrowding so I can assess the impact of capacity constraints.

Acceptance Criteria:

- AC1: Users can set congestion levels for stations or line segments.
- AC2: Congestion introduces travel-time penalties.
- AC3: Visual cues indicate crowding level (heatmap or density indicators).
- AC4: Routing engine favours less-congested paths when appropriate.

Priority: Nice to Have

US-10 – Network Resilience Measurement

As an **analyst**, I want to measure how resilient the network is under disruptions so I can identify critical stations and bottlenecks.

Acceptance Criteria:

- AC1: Average travel time across multiple test routes can be calculated.
- AC2: The system compares metrics before vs after simulated closures.
- AC3: Results can be visualised (charts, colour-coded map, or stats).

- AC4: Users can identify “critical hubs” where closures cause significant impacts.

Priority: Nice to Have

US-11 – Enhanced 3D Visualisation

As a **user**, I want a 3D visual model of the Underground network, so the interface is more intuitive and engaging.

Acceptance Criteria:

- AC1: Stations and lines can be rendered with simple depth/3D shapes.
- AC2: Visual clarity is preserved during rotation, panning, and zooming.
- AC3: Rendering performance remains smooth and responsive.
- AC4: 3D mode does not obscure core map functionality.

Priority: Nice to Have

US-12 – Save & Load Simulation Scenarios

As a **planner**, I want to save and load custom disruption scenarios so I can reuse or share them later.

Acceptance Criteria:

- AC1: Users can save all simulated closures, line suspensions, and congestion states.
- AC2: Scenarios are stored as JSON.
- AC3: Loading a scenario restores all map and routing conditions instantly.
- AC4: Invalid files produce meaningful error messages.

Priority: Nice to Have

US-13 – Route Comparison (Live vs Hypothetical)

As a **researcher**, I want to compare Live and Hypothetical routes so I can analyse the effect of disruptions.

Acceptance Criteria:

- AC1: Dual route results can be displayed side-by-side OR toggled.
- AC2: Differences (e.g., added interchanges, longer routes) are highlighted.
- AC3: Travel-time change (%) is shown.
- AC4: User can export comparison results.

Priority: Nice to Have

US-14 – Export & Reporting Tools

As an **analyst**, I want to export simulation results so I can include them in reports.

Acceptance Criteria:

- AC1: Export available as PDF, PNG, or JSON.
- AC2: Export includes routes, metrics, disruption list, timestamp.
- AC3: Map snapshot included for PDF/PNG exports.
- AC4: Exports produce consistent formatting.

Priority: Nice to Have

US-15 – Scenario Reset Function

As a **user**, I want a Reset option so I can instantly restore the system to the real-time network.

Acceptance Criteria:

- AC1: Reset removes all user-created hypothetical disruptions.
- AC2: Live TfL data is re-applied to the graph.
- AC3: Map updates visually and immediately.
- AC4: User is prompted to confirm before resetting.

Priority: Should Have